Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the f1 score of EfficientNet by dataset.

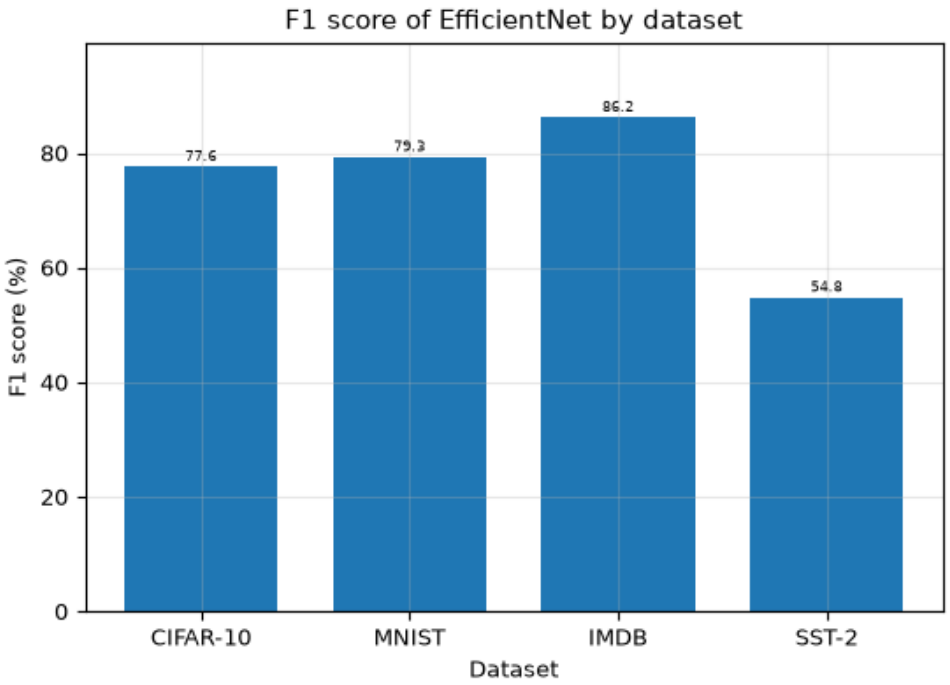


Figure 1: F1 score of EfficientNet by dataset for the models in this experiment.

EfficientNet scores 14.6 points higher on MNIST than on SST-2. On MNIST, EfficientNet obtains a F1 score of 77.6%. On IMDB, EfficientNet obtains a F1 score of 84.2%. As shown in Figure 1, EfficientNet scores 17.8 points higher on CIFAR-10 than on SST-2.

Figure 2 shows the test accuracy during training.

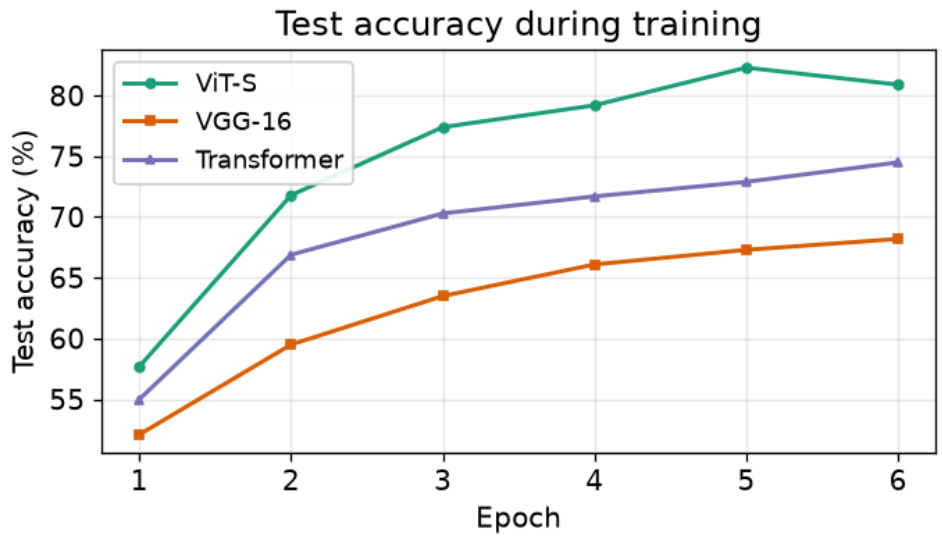


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 6, ViT-S is 13 points better than VGG-16. At epoch 3, VGG-16 achieves

about 60% test accuracy. As shown in Figure 2, Transformer attains a test accuracy of 67% at epoch 2.

3 Discussion

On STL-10, XGBoost obtains an accuracy of 79.0%. The differences between the models are discussed in the next section.